

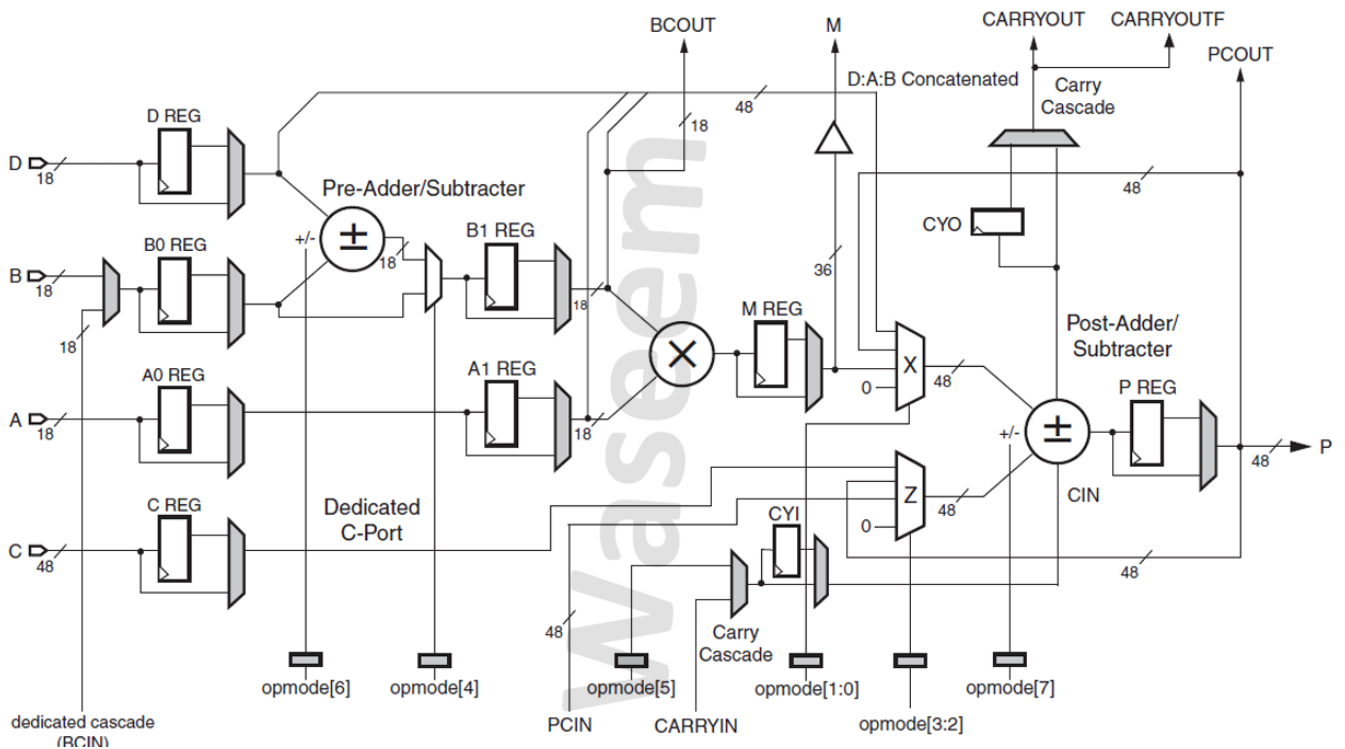
DIGITAL ICS DESIGN

DSP48A1 PROJECT

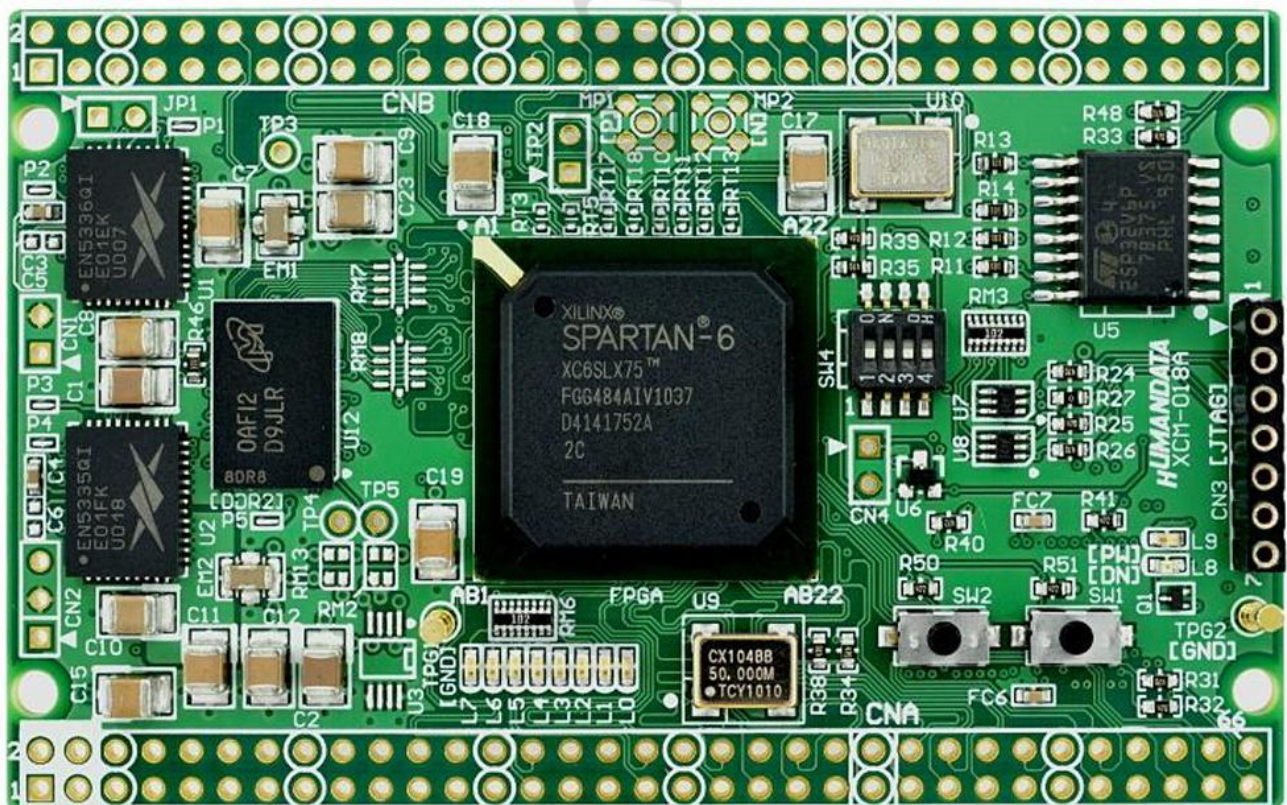


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❖ Design Diagram



❖ The FPGA utilized in the project



RTL code :

➤ MUX_4

```
1  module mux_4(i_sel,i_in0,i_in1,i_in2,i_in3,o_out);
2
3      //inputs
4      input [1:0] i_sel;
5      input [47:0] i_in0,i_in1,i_in2,i_in3;
6
7      //output
8      output reg [47:0] o_out;
9
10     always @(*) begin
11         case (i_sel)
12             2'b00 : o_out = i_in0;
13             2'b01 : o_out = i_in1;
14             2'b10 : o_out = i_in2;
15             2'b11 : o_out = i_in3;
16             default o_out = 48'd0;
17         endcase
18     end
19
20 endmodule
```

➤ DFF with MUX

```
1  module dff_mux(i_clk,i_rst,i_en,i_data,o_data);
2
3      //parameter
4      parameter WIDTH = 18;
5      parameter SEL = 1;
6      parameter RSTTYPE = "SYNC"; //ASYNC or SYNC
7
8      //inputs
9      input i_en,i_clk,i_rst;
10     input [WIDTH-1:0] i_data;
11
12     //output
13     output [WIDTH-1:0] o_data;
14
15     reg [WIDTH-1:0] q;
16
17     generate
18         if (RSTTYPE == "SYNC") begin
19             always @(posedge i_clk) begin
20                 if(i_rst)
21                     q <= 0;
22                 else begin
23                     if(i_en)
24                         q <= i_data;
25                 end
26             end
27         end
28         else if(RSTTYPE == "ASYNC") begin
29             always @(posedge i_clk or posedge i_rst) begin
30                 if(i_rst)
31                     q <= 0;
32                 else begin
33                     if(i_en)
34                         q <= i_data;
35                 end
36             end
37         end
38     endgenerate
39
40     assign o_data =(SEL == 1)? q:i_data;
41
42 endmodule
```

➤ DSP48A1

- Parameters, Inputs, Outputs, wires

```

1  module DSP48A1(A,B,C,D,CLK,CARRYIN,OPMODE,BCIN,
2      RSTA,RSTB,RSTM,RSTP,RSTC,RSTD,RSTCARRYIN,RSTOPMODE,
3      CEA,CEB,CEM,CEP,CEC,CED,CECARRYIN,
4      CEOPMODE,PCIN,BCOUT,PCOUT,P,M,CARRYOUT,CARRYOUTF);
5
6      //===== PARAMETERS =====
7      parameter A0REG = 0; parameter A1REG = 1;
8      parameter B0REG = 0; parameter B1REG = 1;
9      parameter CREG = 1;
10     parameter DREG = 1;
11     parameter MREG = 1;
12     parameter PREG = 1;
13     parameter CARRYINREG = 1;
14     parameter CARRYOUTREG = 1;
15     parameter OPMODEREG = 1;
16     parameter CARRYINSEL = "OPMODE5"; //OPMODE5 OR CARRYIN
17     parameter B_INPUT = "DIRECT"; //DIRECT OR CASCADE
18     parameter RSTTYPE = "SYNC"; //SYNC OR ASYNC
19     //=====
20
21     //===== INPUTS =====
22     input CEA,CEB,CEM,CEP,CEC,CED,CECARRYIN,CEOPMODE;
23     input RSTA,RSTB,RSTM,RSTP,RSTC,RSTD,RSTCARRYIN,RSTOPMODE;
24     input CLK;
25     input CARRYIN;
26     input [7:0]OPMODE;
27     input [17:0] A,B,D,BCIN;
28     input [47:0] C,PCIN;
29     //=====
30
31     //===== OUTPUTS =====
32     output CARRYOUT,CARRYOUTF;
33     output [17:0] BCOUT;
34     output [35:0] M;
35     output [47:0] P,PCOUT;
36     //=====
37
38     //===== WIRES =====
39     wire [17:0] A0_REG,A1_REG,B0_REG,B1_REG,D_REG,B_out;
40     wire [47:0] C_REG,post_add_sub,out_X,out_Z;
41     wire [35:0] M_REG,out_multiplier;
42     wire [17:0] pre_add_sub,i_B1_REG;
43     wire i_CYI,o_CYI;
44     wire i_CYO;
45     wire o_opmode_4,o_opmode_5,o_opmode_6,o_opmode_7;
46     wire [1:0] o_opmode_0_1,o_opmode_3_2;
47     //=====

```


- *Instantiation DFF with MUX*

```

50 //===== INSTANTIATIONS DFF_MUX =====
51 dff_mux #(.WIDTH(18),.SEL(A0REG),.RSTTYPE(RSTTYPE)) a0reg(.i_clk(CLK),.i_rst(RSTA),.i_en(CEA),.i_data(A),.o_data(A0_REG));
52 dff_mux #(.WIDTH(18),.SEL(A1REG),.RSTTYPE(RSTTYPE)) a1reg(.i_clk(CLK),.i_rst(RSTA),.i_en(CEA),.i_data(A0_REG),.o_data(A1_REG));
53 dff_mux #(.WIDTH(18),.SEL(B0REG),.RSTTYPE(RSTTYPE)) b0reg(.i_clk(CLK),.i_rst(RSTB),.i_en(CEB),.i_data(B_out),.o_data(B0_REG));
54 dff_mux #(.WIDTH(18),.SEL(B1REG),.RSTTYPE(RSTTYPE)) b1reg(.i_clk(CLK),.i_rst(RSTB),.i_en(CEB),.i_data(i_B1_REG),.o_data(B1_REG));
55 dff_mux #(.WIDTH(18),.SEL(DREG),.RSTTYPE(RSTTYPE)) dreg(.i_clk(CLK),.i_rst(RSTD),.i_en(CED),.i_data(D),.o_data(D_REG));
56 dff_mux #(.WIDTH(48),.SEL(CREG),.RSTTYPE(RSTTYPE)) creg(.i_clk(CLK),.i_rst(RSTC),.i_en(CEC),.i_data(C),.o_data(C_REG));
57 dff_mux #(.WIDTH(36),.SEL(MREG),.RSTTYPE(RSTTYPE)) mreg(.i_clk(CLK),.i_rst(RSTM),.i_en(CEM),.i_data(out_multiplier),.o_data(M_REG));
58 dff_mux #(.WIDTH(1),.SEL(CARRYINREG),.RSTTYPE(RSTTYPE)) CYI(.i_clk(CLK),.i_rst(RSTCARRYIN),.i_en(CECARRYIN),.i_data(i_CYI),.o_data(o_CYI));
59 dff_mux #(.WIDTH(1),.SEL(CARRYOUTREG),.RSTTYPE(RSTTYPE)) CYO(.i_clk(CLK),.i_rst(RSTCARRYIN),.i_en(CECARRYIN),.i_data(i_CYO),.o_data(CARRYOUT));
60 dff_mux #(.WIDTH(48),.SEL(PREG),.RSTTYPE(RSTTYPE)) preg(.i_clk(CLK),.i_rst(RSTP),.i_en(CEP),.i_data(post_add_sub),.o_data(P));
61
62 dff_mux #(.WIDTH(2),.SEL(OPMODEREG),.RSTTYPE(RSTTYPE)) opmode01(.i_clk(CLK),.i_rst(RSTOPMODE),.i_en(CEOPMODE),.i_data(OPMODE[1:0]),.o_data(o_opmode_0_1));
63 dff_mux #(.WIDTH(2),.SEL(OPMODEREG),.RSTTYPE(RSTTYPE)) opmode32(.i_clk(CLK),.i_rst(RSTOPMODE),.i_en(CEOPMODE),.i_data(OPMODE[3:2]),.o_data(o_opmode_3_2));
64 dff_mux #(.WIDTH(1),.SEL(OPMODEREG),.RSTTYPE(RSTTYPE)) opmode4(.i_clk(CLK),.i_rst(RSTOPMODE),.i_en(CEOPMODE),.i_data(OPMODE[4]),.o_data(o_opmode_4));
65 dff_mux #(.WIDTH(1),.SEL(OPMODEREG),.RSTTYPE(RSTTYPE)) opmode5(.i_clk(CLK),.i_rst(RSTOPMODE),.i_en(CEOPMODE),.i_data(OPMODE[5]),.o_data(o_opmode_5));
66 dff_mux #(.WIDTH(1),.SEL(OPMODEREG),.RSTTYPE(RSTTYPE)) opmode6(.i_clk(CLK),.i_rst(RSTOPMODE),.i_en(CEOPMODE),.i_data(OPMODE[6]),.o_data(o_opmode_6));
67 dff_mux #(.WIDTH(1),.SEL(OPMODEREG),.RSTTYPE(RSTTYPE)) opmode7(.i_clk(CLK),.i_rst(RSTOPMODE),.i_en(CEOPMODE),.i_data(OPMODE[7]),.o_data(o_opmode_7));
68 //=====

```

- *Instantiation MUX_4*

```

70 //===== INSTANTIATION MUX_4 =====
71 mux_4 X(.i_sel(o_opmode_0_1),.i_in0(48'd0),.i_in1({12'd0,M_REG}),.i_in2(P),.i_in3({D[11:0],A1_REG,B1_REG}),.o_out(out_X));
72 mux_4 Z(.i_sel(o_opmode_3_2),.i_in0(48'd0),.i_in1(PCIN),.i_in2(P),.i_in3(C_REG),.o_out(out_Z));
73 //=====

```

- *Assign*

```

75 //===== assign =====
76 assign B_out = (B_INPUT == "DIRECT")?B:(B_INPUT == " CASCADE")?BCIN:0;
77 assign pre_add_sub = (o_opmode_6)?(D_REG-B0_REG) : (D_REG+B0_REG) ;
78 assign i_B1_REG = (o_opmode_4)?pre_add_sub:B0_REG;
79 assign out_multiplier = B1_REG * A1_REG;
80 assign i_CYI = (CARRYINSEL == "OPMODE5")?o_opmode_5:(CARRYINSEL == "CARRYIN")?CARRYIN:0;
81 assign {i_CYO,post_add_sub} = (o_opmode_7)?(out_Z-(out_X+o_CYI)):(out_X+out_Z+o_CYI);
82 assign BCOUF = B1_REG;
83 assign M = M_REG;
84 assign CARRYOUTF = CARRYOUT;
85 assign PCOUT = P;
86 //=====
87
88 endmodule

```

❖ Testbench code

- Inputs and Output

```

1  module DSP48A1_tb();
2
3      //===== INPUTS =====
4      reg CEA,CEB,CEM,CEP,CEC,CED,CECARRYIN,CEOPMODE;
5      reg RSTA,RSTB,RSTM,RSTP,RSTC,RSTD,RSTCARRYIN,RSTOPMODE;
6      reg CLK;
7      reg CARRYIN;
8      reg [7:0]OPMODE;
9      reg [17:0] A,B,D,BCIN;
10     reg [47:0] C,PCIN;
11     //=====
12
13     //===== OUTPUTS =====
14     wire CARRYOUT,CARRYOUTF;
15     wire [17:0] BCOUT;
16     wire [35:0] M;
17     wire [47:0] P,PCOUT;
18     //=====

```

- Instantiation DUT

```

20 // ===== Instantiation DUT =====
21 DSP48A1 #(.A0REG(0),.A1REG(1),.B0REG(0),.B1REG(1),.CREG(1),.DREG(1),.MREG(1),.PREG(1),.CARRYINREG(1),
22     .CARRYOUTREG(1),.OPMODEREG(1),.CARRYINSEL("OPMODE5"),.B_INPUT("DIRECT"),.RSTTYPE("SYNC"))
23     DUT(A,B,C,D,CLK,CARRYIN,OPMODE,BCIN,
24     RSTA,RSTB,RSTM,RSTP,RSTC,RSTD,RSTCARRYIN,RSTOPMODE,
25     CEA,CEB,CEM,CEP,CEC,CED,CECARRYIN,
26     CEOPMODE,PCIN,BCOUT,PCOUT,P,M,CARRYOUT,CARRYOUTF);
27 // =====

```

- Clock generation

```

29 // ===== clock generation =====
30 initial begin
31     CLK = 0;
32     forever #1 CLK = ~CLK;
33 end
34 // =====

```

- Stimulus generations

2.1 Verify Reset Operation

```

36 // ===== Stimulus Generation =====
37 initial begin
38     //2.1 Verify Reset Operation
39     {RSTA,RSTB,RSTM,RSTP,RSTC,RSTD,RSTCARRYIN,RSTOPMODE} = {8'b1111_1111};
40     {CEA,CEB,CEM,CEP,CEC,CED,CECARRYIN,CEOPMODE} = {8'b0101_0010};
41     CARRYIN = $random;
42     OPMODE = $random;
43     A = $random;
44     B = $random;
45     D = $random;
46     BCIN = $random;
47     C = $random;
48     PCIN = $random;
49     @(negedge CLK);
50     if (CARRYOUT != 1'b0 || CARRYOUTF != 1'b0 || BCOUT != 18'd0 || M != 36'd0 || P != 48'd0 || PCOUT != 48'd0) begin
51         $display("Time=%0t,CARRYOUT = %b,CARRYOUTF = %b,BCOUT = %h,M = %b,P = %h,PCOUT = %h", $time,CARRYOUT,CARRYOUTF,BCOUT,M,P,PCOUT);
52         $stop;
53     end
54     {RSTA,RSTB,RSTM,RSTP,RSTC,RSTD,RSTCARRYIN,RSTOPMODE} = {8'b0000_0000};
55     {CEA,CEB,CEM,CEP,CEC,CED,CECARRYIN,CEOPMODE} = {8'b1111_1111};

```

2.2. Verify DSP Path 1

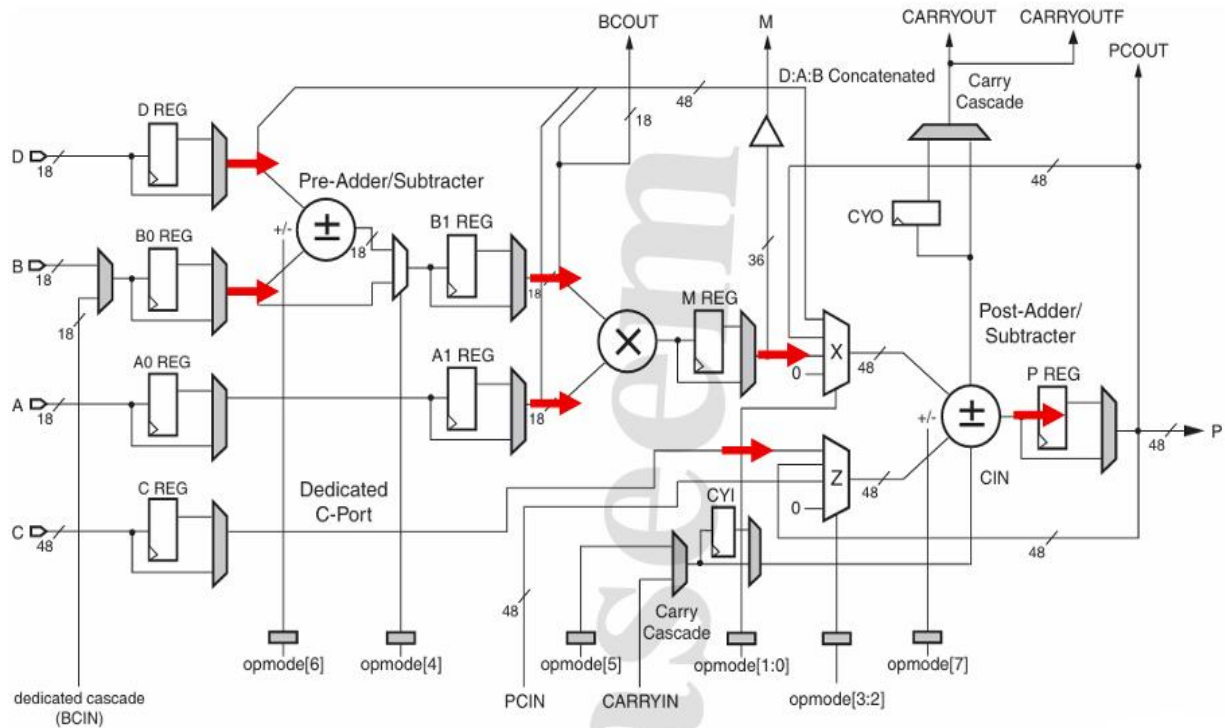


Figure 1: Path 1 Data Flow

```

57 //2.2. Verify DSP Path 1
58 OPMODE = 8'b1101_1101;
59 A = 18'd20; B = 18'd10; C = 48'd350 ; D = 18'd25;
60 BCIN = $random;
61 CARRYIN = $random;
62 PCIN = $random;
63 repeat(4) @(negedge CLK);
64 if (CARRYOUT != 1'b0 || CARRYOUTF != 1'b0 || BCOUT != 18'hf || M != 36'h12c || P != 48'h32 || PCOUT != 48'h32) begin
65     $display( "Time=%0t,CARRYOUT =%b,CARRYOUTF =%b,BCOUT =%h,M =%h,P=%h,PCOUT=%h", $time,CARRYOUT,CARRYOUTF,BCOUT,M,P,PCOUT);
66     $stop;
67 end

```


2.3. Verify DSP Path 2

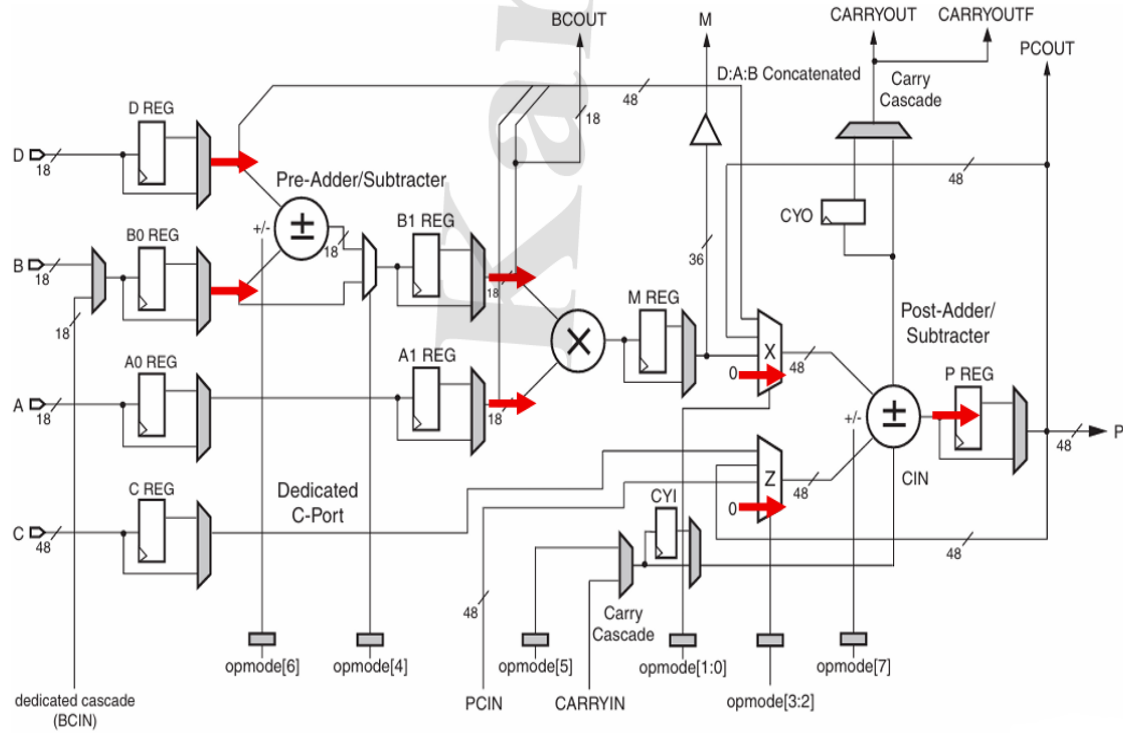


Figure 2: Path 2 Data Flow

```

69 //2.3. Verify DSP Path 2
70 OPMODE = 8'b0001_0000;
71 A = 18'd20; B = 18'd10; C = 48'd350 ; D = 18'd25;
72 BCIN = $random;
73 CARRYIN = $random;
74 PCIN = $random;
75 repeat(3) @(negedge CLK);
76 if (CARRYOUT != 1'b0 || CARRYOUTF != 1'b0 || BCOUT != 18'h23 || M != 36'h2bc || P != 48'd0 || PCOUT != 48'd0) begin
77     $display( "Time=%0t,CARRYOUT =%b,CARRYOUTF =%b,BCOUT =%h,M =%h,P=%h,PCOUT=%h", $time,CARRYOUT,CARRYOUTF,BCOUT,M,P,PCOUT);
78     $stop;
79 end

```

2.4. Verify DSP Path 3

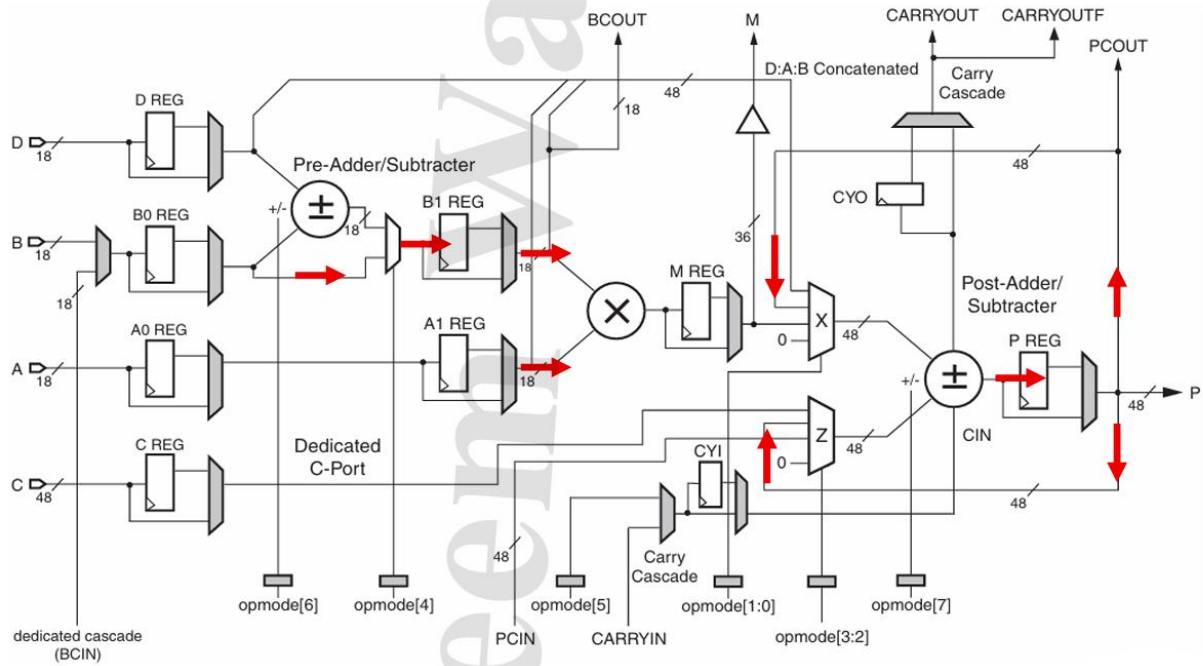


Figure 3: Path 3 Data Flow

```

81 //2.4. Verify DSP Path 3
82 OPMODE = 8'b0000_1010;
83 A = 18'd20; B = 18'd10; C = 48'd350 ; D = 18'd25;
84 BCIN = $random;
85 CARRYIN = $random;
86 PCIN = $random;
87 repeat(3) @(negedge CLK);
88 if (CARRYOUT != 1'b0 || CARRYOUTF != 1'b0 || BCOUT != 18'ha || M != 36'hc8 || P != 48'd0 || PCOUT != 48'd0) begin
89     $display( "Time=%0t,CARRYOUT =%b,CARRYOUTF =%b,BCOUT =%h,M =%h,P=%h,PCOUT=%h", $time,CARRYOUT,CARRYOUTF,BCOUT,M,P,PCOUT);
90     $stop;
91 end

```

2.5. Verify DSP Path 4

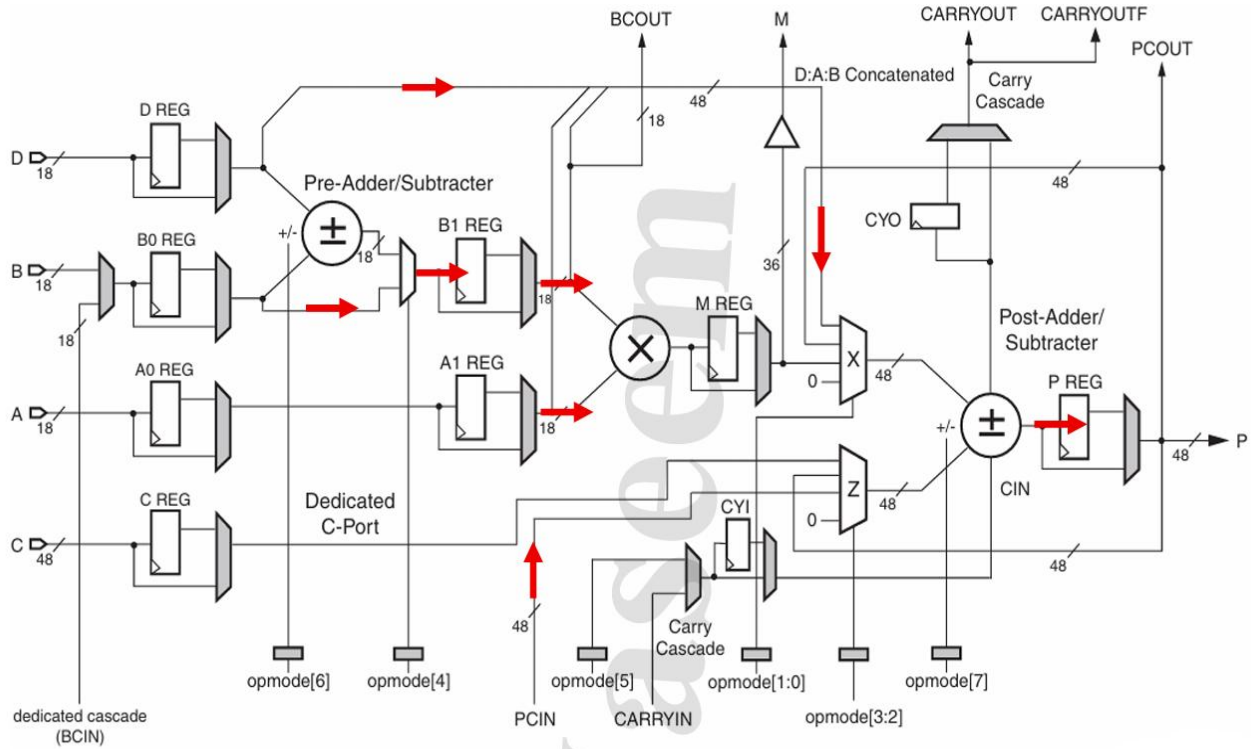


Figure 4: Path 4 Data Flow

```

93 //2.5. Verify DSP Path 4
94 OPMODE = 8'b1010_0111;
95 A = 18'd5; B = 18'd6; C = 48'd350; D = 18'd25; PCIN = 48'd3000;
96 BCIN = $random;
97 CARRYIN = $random;
98 repeat(3) @(negedge CLK);
99 if (CARRYOUT != 1'b1 || CARRYOUTF != 1'b1 || BCOUT != 18'h6 || M != 36'h1e || P != 48'hfe6fffec0bb1 || PCOUT != 48'hfe6fffec0bb1) begin
100     $display( "Time=%0t,CARRYOUT =%b,CARRYOUTF =%b,BCOUT =%h,M =%h,P=%h,PCOUT=%h", $time,CARRYOUT,CARRYOUTF,BCOUT,M,P,PCOUT);
101     $stop;
102 end
103 $stop;
104 end
105 // =====
106
107 endmodule

```

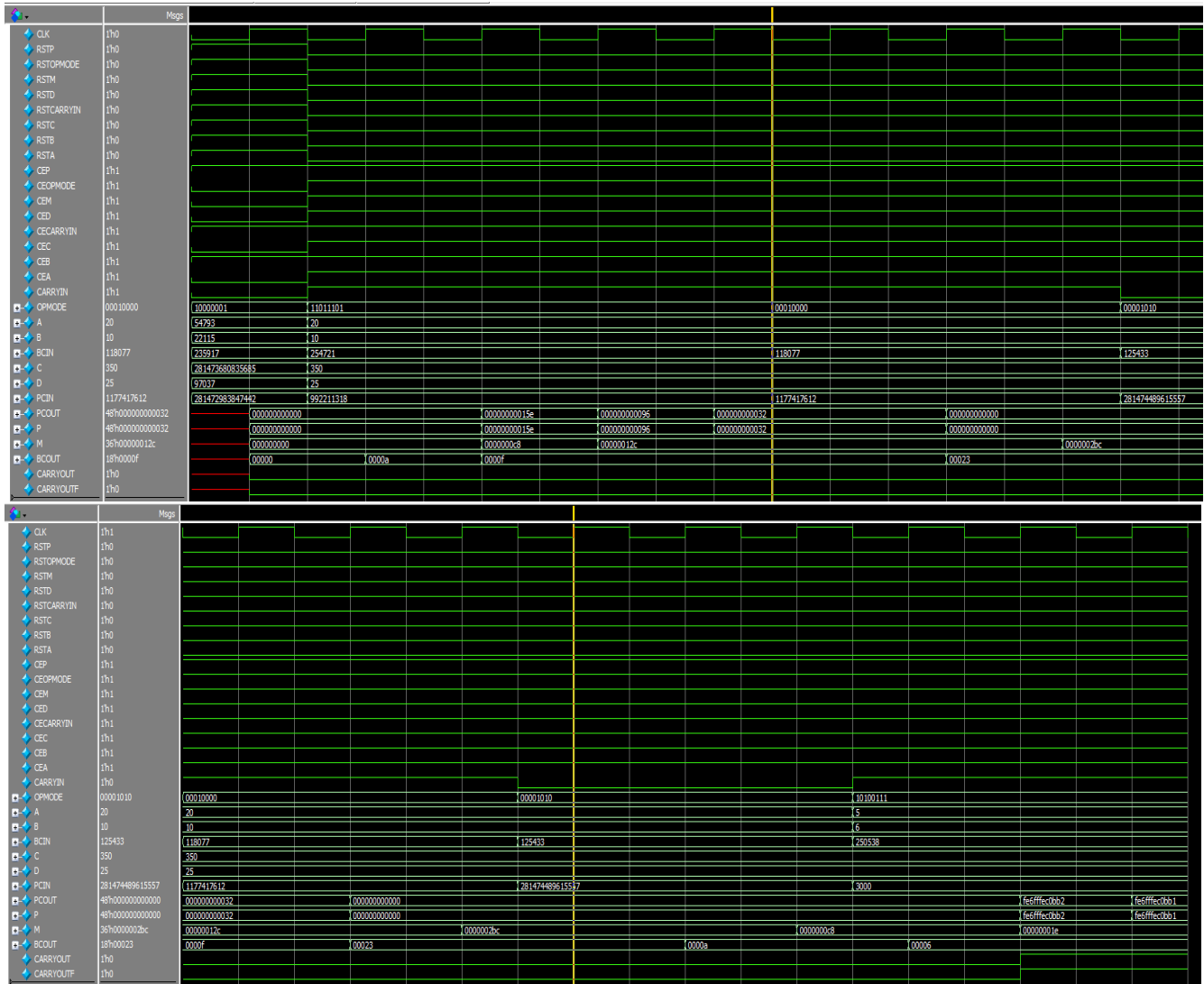
❖ DO FILE

```

run.do
1  vlib work
2  vlog DSP48A1.v DSP48A1_tb.v
3  vsim -voptargs=+acc work.DSP48A1_tb
4  add wave -r *
5  run -all
6  #quit -sim

```

❖ QuestaSim Snippets



❖ Constrain File

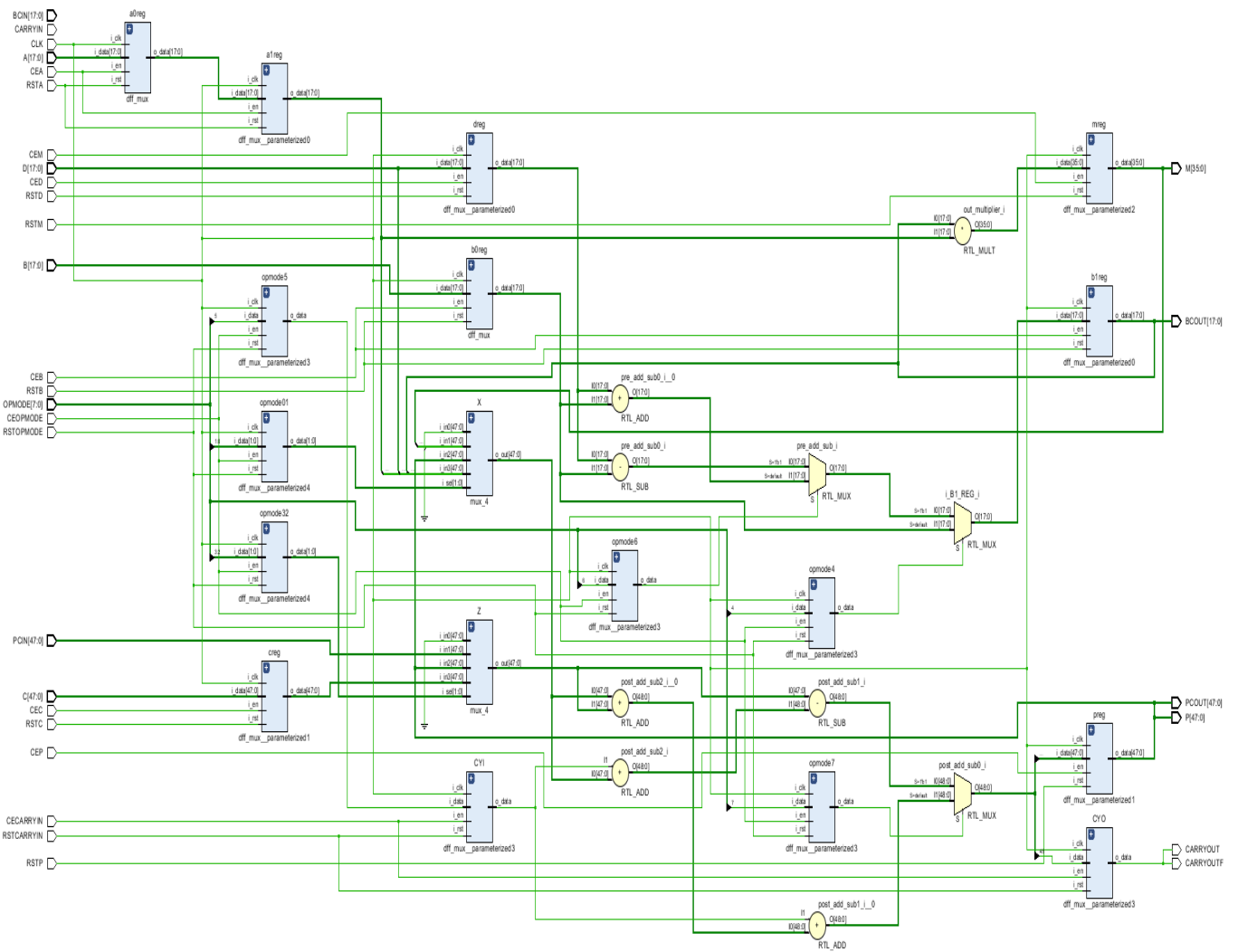
```
1 set_property -dict { PACKAGE_PIN W5 IOSTANDARD LVCMOS33 } [get_ports CLK]
2 create_clock -add -name sys_clk_pin -period 10.000 -waveform {0 5} [get_ports CLK]
```

❖ Elaboration

➤ Messages tab

- ❏ Vivado Commands (3 infos, 4 status messages)
 - ❏ General Messages (3 infos, 4 status messages)
 - ❗ [IP_Flow 19-234] Refreshing IP repositories
 - ❗ [IP_Flow 19-1704] No user IP repositories specified
 - ❗ [IP_Flow 19-2313] Loaded Vivado IP repository 'C:/Xilinx/Vivado/2018.2/data/ip'.
 - ⓘ Command: synth_design -rtl -name rtl_1 (3 more like this)
 - ❏ Elaborated Design (2 infos, 5 status messages)
 - ❏ General Messages (2 infos, 5 status messages)
 - ❗ [Project 1-570] Preparing netlist for logic optimization
 - ❏ Processing XDC Constraints (4 more like this)
 - ❗ Initializing timing engine
 - ❗ Parsing XDC File [\[constrain.xdc\]](#)
 - ❗ Finished Parsing XDC File [\[constrain.xdc\]](#)
 - ❗ Completed Processing XDC Constraints
 - ❗ [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).

➤ Schematic snippets



❖ Synthesis

➤ *Messages tab*

Tcl Console Messages x Log Reports Design Runs

[?] [Info (44)] [Status (19)] Show All

Vivado Commands (3 infos, 6 status messages)

- General Messages (3 infos, 6 status messages)

Synthesis (35 infos, 11 status messages)

- Command: synth_design -top DSP48A1-part xc7a200ffg1156-3 (10 more like this)
 - [Common 17-349] Got license for feature 'Synthesis' and/or device 'xc7a200f'
 - [Synth 8-6157] synthesizing module 'DSP48A1' [DSP48A1.v.1] (7 more like this)
 - [Synth 8-6155] done synthesizing module 'dff_mux' [tff] [dff_mux.v.1] (7 more like this)
 - [Synth 8-226] default block is never used [mux_4.v.1]
 - [Device 21-403] Loading part xc7a200ffg1156-3
 - [Project 1-238] Implementation specific constraints were found while reading constraint file [E:/Assignments_FPGA_wassim/Project_DSP48A1/constrain.xdc]. These constraints will be ignored for synthesis but will be used in implementation. Impacted constraints are listed in the file [X:/DSP48A1_promplm.xdc].
Resolution: To avoid this warning, move constraints listed in [Undefined] to another XDC file and exclude this new file from synthesis with the used_in_synthesis property (File Properties dialog in GUI) and re-run elaboration/synthesis.
- [Synth 8-5818] HDI_ADVISOR - The operator resource <adder> is shared. To prevent sharing consider applying a KEEP on the output of the operator [DSP48A1.v.77] (1 more like this)
- [Synth 8-5842] Cannot pack DSP OPIMODE registers because of constant '1' value. Packing the registers will cause simulation mismatch at initial cycle [dff_mux.v.21]
- [Project 1-571] Translating synthesized netlist
- [Netlist 29-17] Analyzing 207 Unisim elements for replacement
- [Netlist 29-28] Unisim Transformation completed in 0 CPU seconds
- [Project 1-570] Preparing netlist for logic optimization (1 more like this)
- [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).
- [Project 1-111] Unisim Transformation Summary:
No Unisim elements were transformed. (1 more like this)
- [Common 17-83] Releasing license: Synthesis
- [Common 17-1381] The checkpoint E:/Assignments_FPGA_wassim/Project_DSP48A1/project_1/project_1_runs/synth_1/DSP48A1.dcp has been generated.
- [runtcd-4] Executing : report_utilization -file DSP48A1_utilization_synth.rpt -pb DSP48A1_utilization_synth.pb
- [Common 17-206] Exiting Vivado at Tue Jul 14 20:32:06 2026..

Synthesized Design (6 infos, 2 status messages)

- General Messages (6 infos, 2 status messages)
 - [Netlist 29-17] Analyzing 207 Unisim elements for replacement
 - [Netlist 29-28] Unisim Transformation completed in 0 CPU seconds
 - [Project 1-479] Netlist was created with Vivado 2018.2
 - [Project 1-570] Preparing netlist for logic optimization
- Parsing XDC File [constrain.xdc] (1 more like this)
 - [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).
 - [Project 1-111] Unisim Transformation Summary:
No Unisim elements were transformed.

Tcl Console Messages Log Reports Design Runs

Warning (43) Info (44) Status (19) Show All

Synthesis (43 warnings)

- [Synth 8-2490] overwriting previous definition of module DSP48A1 [DSP48A1.v:1]
- [Synth 8-3331] design dff_mux has unconnected port i_clk (40 more like this)
 - [Synth 8-3331] design dff_mux has unconnected port i_rst
 - [Synth 8-3331] design dff_mux has unconnected port i_en
 - [Synth 8-3331] design DSP48A1 has unconnected port CARRYIN
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[17]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[16]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[15]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[14]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[13]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[12]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[11]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[10]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[9]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[8]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[7]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[6]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[5]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[4]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[3]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[2]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[1]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[0]
 - [Synth 8-3331] design DSP48A1 has unconnected port CARRYIN
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[17]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[16]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[15]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[14]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[13]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[12]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[11]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[10]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[9]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[8]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[7]
 - [Synth 8-3331] design DSP48A1 has unconnected port BCIN[6]

➤ Utilization report

Name	Slice LUTs (134600)	Slice Registers (269200)	DSP s (740)	Bonded IOB (500)	BUFGCTRL (32)
▼ N DSP48A1	231	160	1	327	1
a1reg (dff_mux__para...	0	18	0	0	0
b1reg (dff_mux__para...	0	18	0	0	0
creg (dff_mux__param...	0	48	0	0	0
CYI (dff_mux__parame...	1	1	0	0	0
CYO (dff_mux__param...	0	1	0	0	0
dreg (dff_mux__param...	18	18	0	0	0
mreg (dff_mux__para...	1	0	1	0	0
opmode4 (dff_mux__p...	0	1	0	0	0
opmode5 (dff_mux__p...	0	1	0	0	0
opmode6 (dff_mux__p...	17	1	0	0	0
opmode7 (dff_mux__p...	1	1	0	0	0
opmode01 (dff_mux__...	144	2	0	0	0
opmode32 (dff_mux__...	49	2	0	0	0
preg (dff_mux__param...	0	48	0	0	0

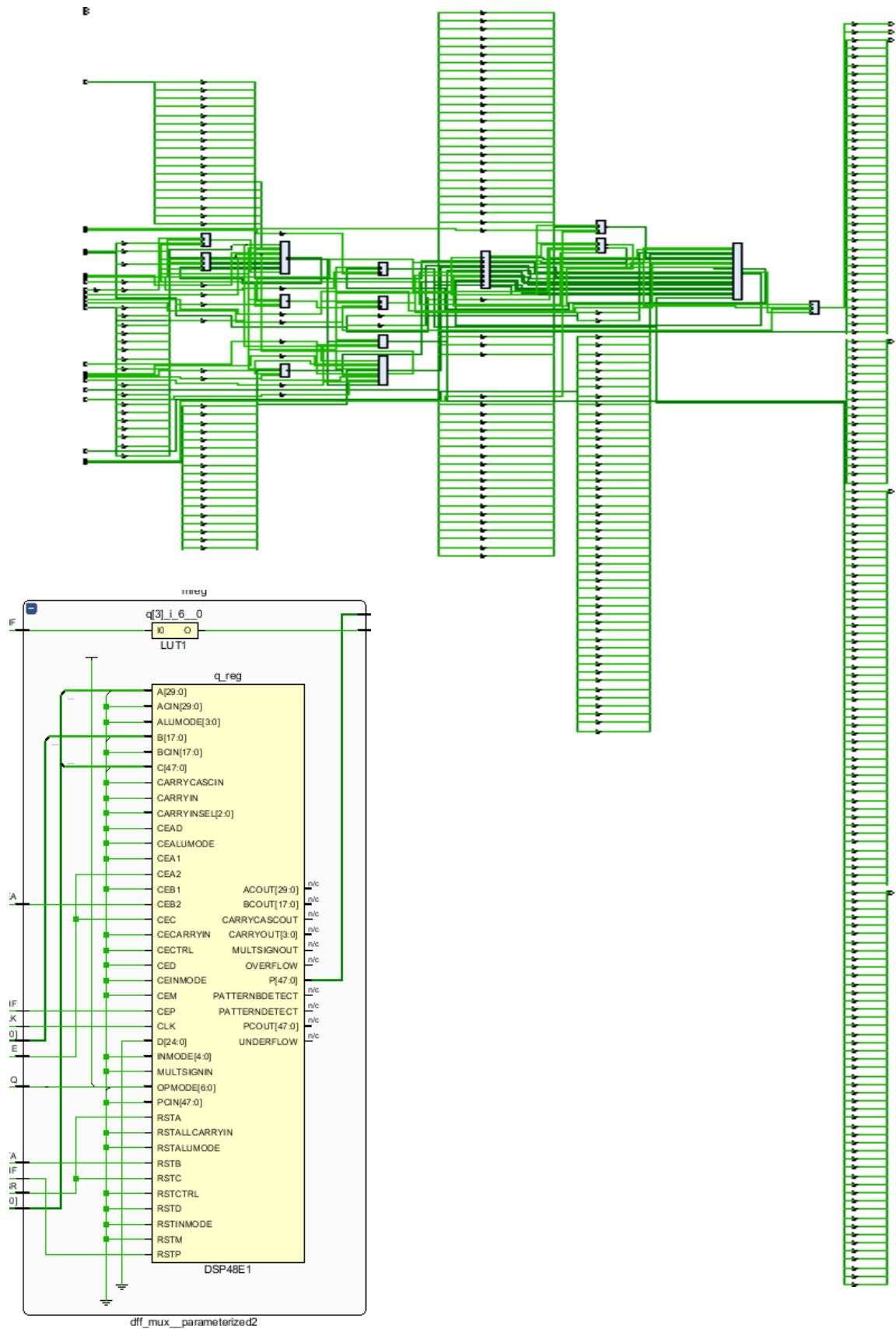
➤ timing report

Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 5.925 ns	Worst Hold Slack (WHS): 0.182 ns	Worst Pulse Width Slack (WPWS): 4.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 106	Total Number of Endpoints: 106	Total Number of Endpoints: 162

All user specified timing constraints are met.

➤ Schematic snippets



➤ Utilization report

Name	Slice LUTs (133800)	Slice Registers (267600)	Slice (33450)	LUT as Logic (133800)	LUT Flip Flop Pairs (133800)	DSP s (740)	Bonded IOB (500)	BUFGCTRL (32)
▼ N DSP48A1	230	179	93	230	52	1	327	1
a1reg (dff_mux__para...	0	18	8	0	0	0	0	0
b1reg (dff_mux__para...	0	36	10	0	0	0	0	0
creg (dff_mux__param...	0	48	15	0	0	0	0	0
CYI (dff_mux__parame...	1	1	1	1	1	0	0	0
CYO (dff_mux__param...	0	2	2	0	0	0	0	0
dreg (dff_mux__param...	18	18	15	18	0	0	0	0
mreg (dff_mux__para...	0	0	0	0	0	1	0	0
opmode4 (dff_mux__p...	0	1	1	0	0	0	0	0
opmode5 (dff_mux__p...	0	1	1	0	0	0	0	0
opmode6 (dff_mux__p...	17	1	6	17	0	0	0	0
opmode7 (dff_mux__p...	1	1	2	1	0	0	0	0
opmode01 (dff_mux__...	144	2	44	144	0	0	0	0
opmode32 (dff_mux__...	49	2	30	49	0	0	0	0
preg (dff_mux__param...	0	48	12	0	0	0	0	0

➤ timing report

Design Timing Summary

Setup

Worst Negative Slack (WNS): 4.827 ns
 Total Negative Slack (TNS): 0.000 ns
 Number of Failing Endpoints: 0
 Total Number of Endpoints: 125

Hold

Worst Hold Slack (WHS): 0.139 ns
 Total Hold Slack (THS): 0.000 ns
 Number of Failing Endpoints: 0
 Total Number of Endpoints: 125

Pulse Width

Worst Pulse Width Slack (WPWS): 4.500 ns
 Total Pulse Width Negative Slack (TPWS): 0.000 ns
 Number of Failing Endpoints: 0
 Total Number of Endpoints: 181

All user specified timing constraints are met.

➤ device snippets

